

Christopher (Christy) Bruce

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EDUCATION

Loughborough University

MSc. Financial Mathematics | Distinction (**80.5%**, Course Rank: 2nd, Scholarship)

BSc. Mathematics Major (Natural Sciences) | First-class honours.

Loughborough, UK

Sep 2023 – Oct 2024

Sep 2019 – Jun 2022

Relevant Modules: Statistics for Large Data (Data Science, XGBoost, Random Forests) (83%),
Timeseries Econometrics (95%), Economic Data Analysis (90%), Probability Theory (79%),
Derivative Pricing (73%), Financial Economics (76%), Computational Finance (83%),

Thesis: Financial Complex Systems & Networks, observing dynamic correlation relationships over time (Scored 93%)

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Statsmodels, Scikit-Learn), R, SQL.

Quantitative/Statistical: Econometrics, Time-Series Analysis, Machine Learning (XGBoost, Random Forests).

PROFESSIONAL EXPERIENCE

Imperial College London

Research Assistant (Quantitative Researcher)

London

Feb 2026 – Present

- Worked with Imperial's Data, Systems, and Finance Labs, producing equity portfolios, producing 2.8% extra return than the market index annually in the US market.
- Responsible for signal and alpha research from NLP text-derived signals, revealing key drivers of excess returns.
- Exercised ML techniques from regressions to XGBoost to link company actions to profitability metrics, using PCA for feature compression and winsorization for outlier dampening.
- Efficient use of Claude code/ Cursor, building an error detection system to ensure robustness.
- Engineered proprietary investment signals by mapping Sustainable Development Goals (SDGs) and corporate actions (e.g., R&D investments) into quantitative portfolio factors.
- Built automated pipelines in Pandas/SQL to synchronize disparate datasets from multiple sources (e.g. WRDS).
- Collaborating with Nikkei Exchange Japanese analysts to build a sustainability-linked tradable index.
- Executed signal robustness tests using rolling Fama-French 3-factor regressions.
- Built a comprehensive strategy performance suite (e.g. VaR, Rolling Sharpe).

All Options

Quantitative Trader Programme

Amsterdam

Sept 2025 - Oct 2025

- Selected with a 0.5% acceptance rate to the programme, involving an intensive course in **option theory**, along with market-making games and hands-on shadowing of options traders and quantitative researchers.
- Enhanced my ability to tackle complex problems in a fast-changing environment, under pressure.

Introspect Capital

Quantitative Analyst Intern

London

May 2024 - Oct 2024

- **Algorithmic Modelling & Risk Analytics:** Developed an unsupervised learning (Network Clustering) framework to identify hidden market structures, contributing to more accurate risk assessment.
- **Data Manipulation:** Designed SQL and Python pipelines to synchronise and clean factor data for 500+ stocks, ensuring data integrity for back-testing and validating using Pandas and Numpy.
- **Machine Learning:** leveraged XGBoost for predictive modeling of financial time-series, focusing on minimizing model error and understanding feature importance for stakeholder reporting.
- **Communication/Team:** Communicated complex technical findings through data visualization and comprehensive research reports for senior management; collaborated with cross-functional teams.

RESEARCH PROJECTS

Masters Thesis Presented at a Conference (Complex Systems & Networks) (Scored 93%)

Loughborough

Researcher

Jun 2024 – Present

- Built a Python-based algorithm to dynamically track changes in correlation structure in equity markets.
- Analysed ~2,500 stocks over two years to identify how systemic shocks (e.g. Covid-19) altered clustering, risk concentration, and market leadership.
- Used clustering and network metrics to **detect regime shifts** and periods of elevated systemic risk, supporting more informed **risk-adjusted investment decisions**.
- Presented my findings to the NetSci conference with an audience of over 50 professionals, explaining complex topics into clear, actionable insights.

Custom Machine Learning Model from Scratch (Python)

Amsterdam

Researcher

April 2025 – Jan 2026

- Designed and implemented a novel, probabilistic machine learning architecture from first principles to model and forecast complex financial time-series.
- Developed a predictive framework estimating expected values via conditional probabilities derived from historical binary pattern sequences.
- Built an independent, robust out-of-sample backtesting engine; achieved a 2.5 Sharpe ratio over a 6-year historical window on US equity indices.
- Integrated a dynamic volatility-scaling mechanism utilizing GARCH models and model-confidence metrics to optimize risk-adjusted performance.
- Presented the methodology to the Amsterdam Investment Club, winning an award for exceptional creativity.

Prediction Market Systematic Model (Event-Driven) (Python)

Amsterdam

Co-Researcher

Oct 2025 – Jan 2026

- Engineered an event-driven statistical model to analyze event-contract markets, deriving implied probability distributions and evaluating deviation from market consensus.
- Developed automated API pipelines using Claude code for real-time data ingestion, cleaning, and state-tracking across fragmented data sources.
- Backtested and deployed the model live, achieving a top 5% performance rank on Polymarket, modelling mispriced probabilities.

IMC Prosperity, Algorithmic Trading Challenge

London

Co-Researcher

April 2025 – May 2025

- Placed in the top 3% out of 12,000+ global teams in an intensive, multi-round algorithmic simulation competing against international quantitative talents.
- Implemented a mathematical Black-Scholes pricing framework to value derivative contracts, executing relative-value trading strategies based on structural mispricings.
- Created arbitrage strategies between assets, managing risk, e.g. by dynamically adjusting position size.

LEADERSHIP & EXTRACURRICULAR

Amsterdam Sahaj Yoga Meditation Course

Amsterdam

Co-Organiser

Oct 2025 – Feb 2026

- Lead various meditation workshops weekly, giving presentations on topics related to meditation and balance
- Guiding participants one-to-one for balance and a clear mind through meditation.

International Finance Conference

French Alps

Co-Organiser

Jan 2024 – Jun 2024

- *Team Leadership/Communication:* Initiated and co-organised a Quant Finance research conference, presenting 4 personal research projects and coordinating 10 attendees.